# 1. Introduction to React JS

## ▼ What is React js?

- React JS is a **Frontend JavaScript Library** used for dynamic User Interface (UI).
- React JS is especially used to create **Single-Page Applications (SPA)**.
- SPA → Only one HTML file, reload only the necessary component.
- It allows developers to create reusable UI components, making the development process faster and more efficient.
- React JS follows a **component-based architecture**.

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## ▼ History to React js

- React.js was developed by **Jordan Walke,** a software engineer at **Facebook.**
- Initially Named as "`FAXJS`".
- He created React in 2011 as an internal tool to improve the performance of Facebook's **News Feed** Section.
- React was officially **released to the public in 2013** as an open-source JavaScript library.
- It is now an **open-source** library used by developers worldwide.
- The latest major version of React as of **April 8, 2026** is React (v19) (specifically v19.2.5 released April 2026).

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## ▼ Why Learn React js?

1. High Demand in the Job Market
2. **Simple to Learn** compared to others
3. **Reusable Components** → write once, use everywhere
4. **Huge Community** → easy solutions

5. **Rich Ecosystem** → supports Tons of libraries, tutorials, videos, and documentation available.

### ▼ `In short` : Learning React.js gives you the power to build fast, scalable, and modern web apps, and opens doors to high paying jobs and real-world development opportunities

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## ▼ Prerequisites to Learn React js

- Before learning React.js, you should have basic knowledge of HTML, CSS,
  and JavaScript.
  1. **HTML (Hyper Text Markup Language)** → The structure of a webpage (e.g., headings, buttons, forms).
  2. **CSS (Cascading Style Sheets)** → Used for styling the webpage (e.g., colors, layouts).
  3. **JavaScript (ES6+)** → Basics of JS.

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## ▼ Mandatory JavaScript Prerequisites for Learning React js

- **You MUST know these JavaScript concepts before starting React:**
  1. **ES6 Concepts** → let & const,  Arrow Functions, template literals, Destructuring (arrays & objects), spread and rest operators, Modules (import/export)
  2. **Array Methods** → map, filter, reduce, forEach
  3. Functions & Callbacks

4. Promises & async/await
5. Fetch API
6. Basic Events and Event Handling
7. Basic Understanding of the DOM

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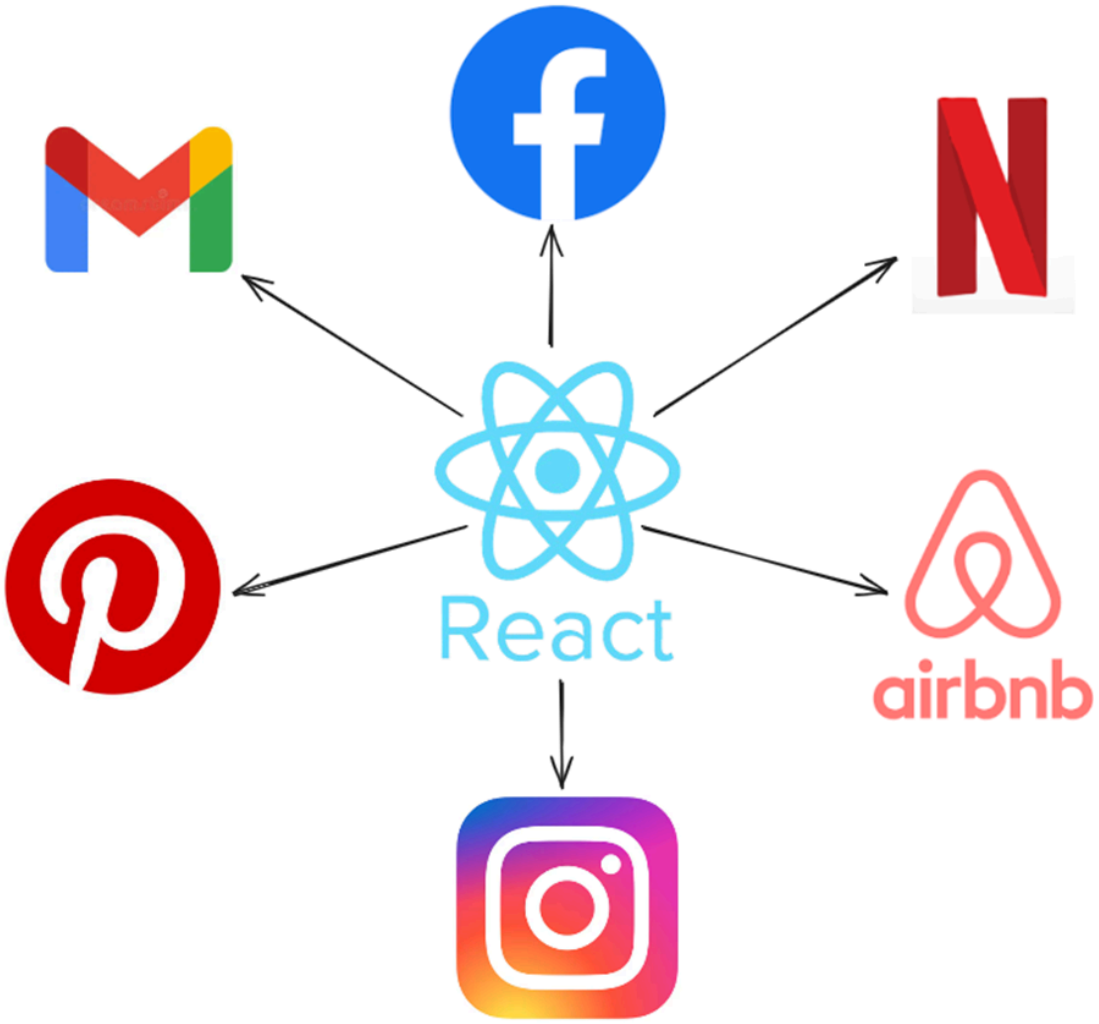
# ▼ List of Companies Using React

- These are some popular companies that use React in their workflow:




- **For Example** → Facebook, Instagram, Gmail, Airbnb, Netflix, Uber, WhatsApp Web, Spotify, etc.

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# ▼ Difference Between Library and Framework

## ▼ What is a Library?

- A **library** is a collection of reusable functions or components that developers can use as needed.
- A library is a collection of reusable code that you can include in your project to help you perform common tasks more easily.
- **For Examples** → React js, Axios, etc.
- **In simple** →
  Imagine you're building a house. A library is like a `toolbox` → you choose the tools you want and use them when needed.

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## ▼ What is a Framework?

- A **framework** is a full-fledged system that provides rules and structure for building an entire application.
- It provides built-in rules, patterns, and features that you must follow.
- **For Examples** → Angular, vue.js, Express.js, Next.js, etc.
- **In Simple** →
  Continuing the house example, a framework is like a `building blueprint` →it tells you how to build the house and in what order.

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# ▼ Difference Between SPA and MPA

| SPA (Single-Page Application) | MPA (Multi-Page Application) |
|---|---|
| A SPA stands for **single-page application**. | A MPA stands for **Multi-page applications**. |
| Loads a **single HTML page** and updates content dynamically without reloading the page. | Each page is a **separate HTML file**, making navigation slower. |
| **Faster after the first load**, as only necessary data is updated. | **Slower performance**, as the browser loads everything again on each page change. |
| Uses **client-side routing** (React Router) to switch views without reloading. | Every new page request **reloads the entire page from the server**. Uses **server-** |

| SPA (Single-Page Application) | MPA (Multi-Page Application) |
| --- | --- |
| | **side routing** (URLs trigger full-page reloads). |
| Examples: **Gmail, Facebook, Twitter, Netflix, YouTube Web App**. | Examples: **Amazon, Wikipedia, News Websites, Banking Portals**. |

# ▼ What is MVC in Angular?

- **MVC** stands for **Model–View**–Controller, a design pattern used to separate
  concerns in application development.
- In Angular, the MVC pattern helps organize the code by separating:
- Data (Model)
- UI (View)
- Logic (Controller)